

Accounting for Data Source Heterogeneity in Ames Mutagenicity QSAR Development and Evaluation

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Abstract

QSAR models for Ames mutagenicity are accepted as supporting evidence under EU REACH, K-REACH, and ICH M7, yet their reported performance varies widely across evaluation protocols, making it difficult to assess their true predictive reliability.¹⁻³ This study systematically evaluated 225 model configurations—six tabular algorithms paired with four feature representations, plus a graph convolutional network with graph features, under three preprocessing scenarios—across three genotoxicity datasets: Ames ($n = 13,560$), in-vitro chromosome aberration ($n = 1,619$), and in-vivo micronucleus ($n = 2,153$). Using a single locked configuration (XGBoost, compact features, raw SMILES) in an evaluation cascade of increasing rigor, Ames MCC fell from 0.670 (random-split CV) to 0.624 (scaffold-split CV) to 0.470 (scaffold-split held-out test) and further to 0.234 and 0.122 (drug→industrial and industrial→drug leave-domain-out, LDO). The split-method effect ($\Delta = 0.046$) was nearly an order of magnitude smaller than the data source heterogeneity effect ($\Delta = 0.390$): a naive source classifier reached $\text{MCC} = 0.608$ without any structural information, reflecting a 19.5-fold prevalence gap between drug-sourced (58.3%) and industrial-sourced (3.0%) compounds. Threshold

decomposition showed that prior shift accounts for roughly one-third (30.9–35.3 %) of the LDO performance gap, leaving a substantial residual that no recalibration can eliminate. In-domain algorithm differences were small (MCC range 0.063), but robustness under source shift was uneven: Random Forest maintained $\text{MCC} = 0.325$ in the hardest direction where XGBoost collapsed to 0.122—suggesting that algorithm choice matters far more under cross-source evaluation than in-domain comparisons indicate. In-vivo micronucleus models showed ranking ability ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) but classification metrics with 95% CIs crossing zero, reflecting both modeling limits and limited test-set positives ($n = 12$). These findings support an evaluation cascade with source-awareness as a useful addition to current reporting practices for mutagenicity QSAR.

Introduction

Under EU REACH⁴ and K-REACH,⁵ industrial chemical manufacturers and importers must assess genotoxic potential as part of their registration dossiers; in silico QSAR predictions are explicitly accepted as supporting evidence when generated by validated models with sufficient applicability domain coverage.⁶

For pharmaceuticals, the ICH M7(R2) guideline plays an analogous role by endorsing statistical QSAR alongside rule-based expert systems for assessing the mutagenic potential of impurities.⁷ Across both regulatory frameworks, the Ames bacterial reverse mutation assay^{8,9} is the central predictive target, with in-vitro chromosome aberration (CA) and in-vivo micronucleus (MN)^{10,11} serving as higher-tier endpoints. Reliable models could reduce animal testing in line with the 3Rs principle.¹²

Published Ames QSAR models report a wide spread of metrics—accuracies of 0.80–0.90 and MCC of 0.60–0.80^{1,13–15}—but these come from different splits, datasets, preprocessing pipelines, and protocols, making direct comparison difficult. At least three factors can inflate apparent performance.

Splitting strategy. Random splits allow structurally similar compounds (sharing the same Murcko scaffold¹⁶) to appear in both training and test set, producing optimistic estimates. Scaffold-based splitting^{3,17} is more realistic, but the magnitude of the random-vs-scaffold gap has not been systematically quantified across genotoxicity endpoints.

Source mixing. Most large Ames datasets aggregate compounds from pharmaceutical libraries and industrial chemical registries,^{1,13} which differ in composition, prevalence, and likely in data quality and labeling conventions. When mixed, models may learn source membership rather than toxicity mechanisms—a form of data source heterogeneity noted^{18,19} but never systematically measured. One component of this heterogeneity is prior shift: the large prevalence gap between sources can miscalibrate the default decision threshold independently of any structural difference. However, threshold decomposition analysis presented here shows that prior shift accounts for only part of the observed performance gap, with a residual that no recalibration can eliminate.

Preprocessing. Salt stripping, metal removal, and charge normalization can substantially alter molecular representations,^{20,21} yet these steps are often undocumented or applied uniformly across endpoints.

Most studies evaluate a single endpoint with a single split, focus on algorithm comparison,^{22,23} rarely test cross-source generalization, and seldom report statistical significance.²⁴ The relative contributions of split method, source composition, and preprocessing have not been disentangled in a single consistent framework.

This study addresses these gaps by:

1. Introducing an evaluation cascade—scaffold-split test, random vs. scaffold CV, and cross-source leave-domain-out—applied consistently to a single locked configuration.
2. Showing that, on the Ames benchmark, data source heterogeneity drives a scaffold-CV-to-LDO MCC drop of 0.390—an order of magnitude larger than the split-method effect ($\Delta = 0.046$)—and that threshold-tuning analysis attributes only part of this drop to prior shift, leaving a residual that no recalibration can eliminate.

3. Finding that in-domain algorithm differences are small (MCC range = 0.063) but that cross-source robustness varies sharply (RF 0.325 vs. XGB 0.122 in the harder direction), so algorithm choice matters substantially more under source shift than in-domain comparisons suggest.
4. Reporting endpoint-specific preprocessing effects, with salt stripping improving the best in-vivo model by +0.22 MCC while harming in-vitro.
5. Releasing an open-source pipeline for full reproducibility.

Materials and Methods

Datasets

Three datasets were evaluated (Table 1): Ames bacterial reverse mutation ($n = 13,560$), in-vitro chromosome aberration ($n = 1,619$), and in-vivo micronucleus ($n = 2,153$).

The Ames dataset combines pharmaceutical (drug-like) and industrial chemical sources. Drug-sourced compounds were obtained from the Hansen benchmark¹³ and the Honma/Furuhamma Ames/QSAR Challenge datasets,^{1,2} which were curated by the original authors using literature-derived Ames assay results without additional Klimisch filtering applied here; these benchmarks are intentionally enriched for known mutagens for evaluation purposes, which contributes to their elevated positive rate. Industrial-sourced compounds were obtained from ECHA REACH dossiers and K-REACH submissions, then filtered by Klimisch reliability score (see below). Compounds carry a source label based on their originating dataset; the term “source” is used here to emphasize that this label reflects data provenance—encompassing potential differences in chemical space, data quality, curation conventions, and pre-existing mutagen enrichment—rather than a computed chemical classification. Drug-sourced compounds ($n = 6,416$) had a positive rate of 58.3%, while industrial-sourced ($n = 7,144$) had 3.0%, a 19.5-fold difference. When a compound appeared in both drug- and

industrial-sourced datasets with consistent labels, the drug-source label was retained (this affected [N] compounds and did not remove any compound from the LDO partition). The in-vitro and in-vivo datasets lacked source annotations.

Raw study records for all three endpoints were first mapped to endpoint categories (Ames, in-vitro CA, in-vivo MN) using a unified classification table spanning OECD, EPA, and EU Test Guidelines; the full guideline-to-endpoint mapping is provided in the supplementary materials. For industrial chemicals sourced from ECHA dossiers, inclusion was restricted to studies with Klimisch reliability scores of 1 (reliable without restriction) or 2 (reliable with restrictions) where available,²⁵ excluding records with score 3 or 4. This Klimisch filter was not applied to the drug-sourced data, since those benchmarks do not carry Klimisch annotations; this asymmetry in quality control should be borne in mind when interpreting the cross-source comparisons. When a single compound had both positive and negative study records of equal Klimisch score, a conservative hazard-assessment convention consistent with EU REACH and K-REACH guidance^{4,5} was applied and the compound was labeled as positive; a sensitivity analysis using the alternative "majority-vote" rule changed the industrial Ames positive rate by less than 0.5 pp and did not alter any reported conclusion (Table S[X]). Compounds with irreconcilable label conflicts across distinct sources were excluded (169 Ames, 3 in-vitro, 2 in-vivo; representing 1.2%, 0.2%, and 0.1% of each raw dataset, respectively).

Cross-endpoint overlap analysis revealed substantial sharing: 97.5% of in-vitro and 92.1% of in-vivo compounds also appeared in Ames, with moderate label concordance (Cohen’s $\kappa = 0.14$ – 0.20).

All datasets were split 80/20 using scaffold-based splitting¹⁷ with zero scaffold-group overlap. Murcko scaffolds¹⁶ were generated with RDKit;²⁶ acyclic compounds were grouped into pseudo-scaffolds via agglomerative clustering with average linkage on 1–Tanimoto similarity (1024-bit Morgan fingerprints,²⁷ distance threshold = 0.6), and each cluster was treated as a single scaffold group during the 80/20 split.

Table 1: Dataset summary. Positive rate refers to the training partition. The 225 total configurations comprise 3 datasets $\times$ 3 preprocessing scenarios $\times$ (6 tabular models $\times$ 4 feature modes + 1 GNN).

Dataset	Assay	n	Train	Test	Pos. rate (%)	Scaffolds
Ames	Bacterial reverse mutation	13,560	10,848	2,712	29.1	1,691
In vitro CA	Chromosome aberration	1,619	1,295	324	10.4	353
In vivo MN	Micronucleus	2,153	1,722	431	3.7 / 2.8 ^a	410

^aTraining / test positive rate.

Preprocessing, Features, and Algorithms

Each dataset was processed under three scenarios: **raw** (original SMILES), **salt-stripped** (largest fragment retained via RDKit’s `LargestFragmentChooser`²⁶ followed by charge normalization—modifying 45% of Ames, 81% of in-vitro, and 83% of in-vivo SMILES), and **no-metal** (compounds with metal atoms excluded).

Five feature representations were evaluated. **Compact** (138 features) combined 13 physicochemical descriptors (MW, LogP, TPSA, HBD, HBA, rotatable bonds, fraction C_{sp3}, ring counts, heteroatoms, heavy atoms, formal charge, valence electrons, aromatic ring count) with 125 structural alerts based on Benigni–Bossa SA1–SA31,^{28,29} Kazius toxicophores,³⁰ and ISS Ames alerts. **Broad FP-512/1024/2048** (266/394/522 features) extended compact with 128/256/384 Morgan fingerprint bits²⁷ selected by mutual information³¹ on the training set only (bits with prevalence below 1% or above 99% excluded). **Graph** representations used molecular graphs with 9-dimensional atom features and normalized adjacency matrices, paired exclusively with the GCN.

Seven classifiers were evaluated: XGBoost (XGB),³² LightGBM (LGBM),³³ Random Forest (RF),³⁴ SVM with RBF kernel,³⁵ Logistic Regression (LR),³⁶ a 128–64 MLP,³⁷ and a three-layer GCN³⁸ with LayerNorm and global mean pooling. SVM and LR used standard scaling. Class imbalance was handled via `scale_pos_weight` or `class_weight='balanced'`. Hyperparameters were tuned via `RandomizedSearchCV` ($n_{\text{iter}} = 50$) with `GroupKFold`³⁹ (5 folds, scaffold groups); the per-algorithm search spaces and selected values for every endpoint

are provided in Table S3.

Evaluation Cascade

For fair comparison across evaluation methods, a single locked configuration—XGB, compact features, raw SMILES—was defined and applied across all three levels of the cascade (Figure 1). XGBoost was chosen as the locked algorithm because gradient boosted trees are the most widely reported family in recent Ames QSAR benchmarks,^{1,2,15} making the cascade results directly comparable to published numbers; the compact feature set and raw SMILES were chosen as the minimum-preprocessing baseline so that the cascade reflects performance without preprocessing- or feature-induced gains. The choice was committed before LDO experiments were run. Sensitivity of the cascade to algorithm choice is examined in Level 3, where LGBM and RF are run alongside XGB.

Level 1 (in-domain test): the scaffold-split held-out test set, with bootstrap 95% CI (500 iterations) on MCC,⁴⁰ ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, and Brier score.⁴¹

Level 2 (split-method comparison): 5-fold stratified random CV and 5-fold scaffold-grouped CV, each with 5 repeats, on the full dataset.

Level 3 (cross-source LDO): for Ames, trained on one source and tested on the other, using the locked configuration plus LGBM and RF to assess algorithm-specific robustness. As in Level 1, MCC and the full panel of secondary metrics (ROC-AUC, balanced accuracy, sensitivity, specificity, Brier score) are reported with 95% bootstrap CIs (500 iterations) on the LDO test partition.

Additional Analyses

Beyond the locked configuration, all 225 combinations were evaluated on the scaffold-split test set ($3 \times 3 \times [6 \times 4 + 1] = 225$). Applicability domain (AD) used Tanimoto similarity on 1024-bit Morgan fingerprints²⁷ with threshold ≥ 0.3 . SHAP values⁴² (**TreeExplainer**)

were computed for tree-based models on the three primary endpoints. Learning curves spanned 10%–100% training fractions with 5-fold scaffold CV under the locked configuration. McNemar’s test⁴³ with continuity correction compared the top two models per endpoint.

LDO threshold decomposition. To disentangle prior shift from genuine source shift, LDO MCC was computed at three thresholds: *default* ($t = 0.5$), *prior-corrected* ($t =$ training positive rate, optimal under pure prior shift), and *oracle* (t maximizing MCC on the test set, an upper bound on threshold-based recalibration).

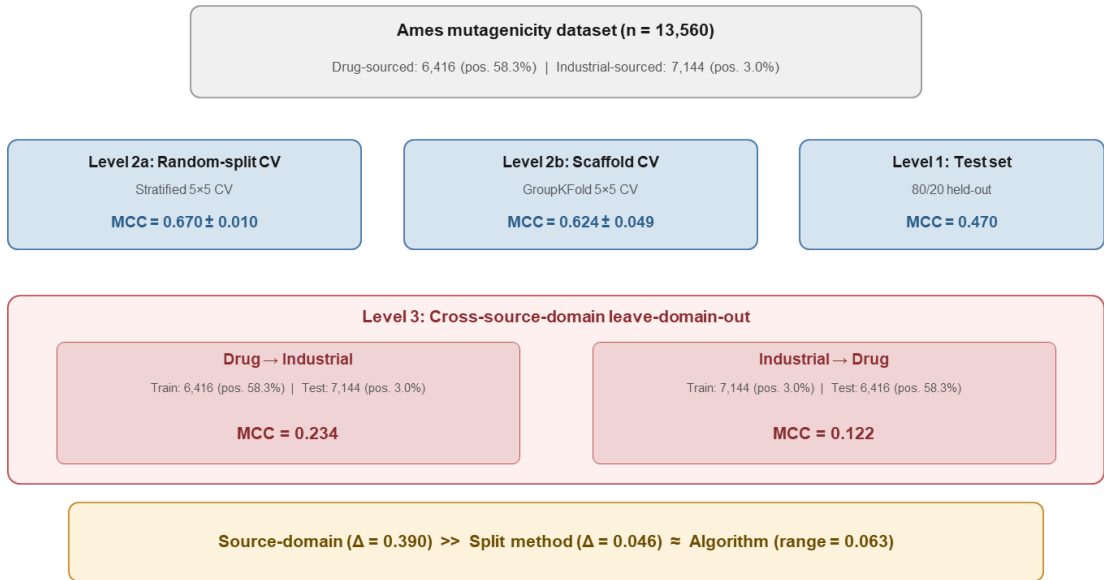


Figure 1: Evaluation cascade framework. All levels use the same locked configuration (XGB, compact features, raw SMILES), enabling fair MCC comparison across evaluation methods.

Results

Best Observed Performance Across the Full Grid

Table 2 reports the best observed performance per endpoint across all 225 configurations. These values are an upper bound from extensive search and should be read as a descriptive

benchmark, not unbiased estimates; the confirmatory analysis uses the locked configuration (Table 4).

Table 2: Best observed performance per endpoint (scaffold-split test set, full grid). Best among 225 configurations; descriptive benchmark, not unbiased estimates.

Endpoint	Algo.	Preproc.	Features (n)	MCC [95% CI]	AUC	PR-AUC	Sens.	Spec.
Ames	XGB	raw	FP-2048 (522)	0.555 [.522,.588]	0.850	0.765	0.756	0.822
In vitro	LGBM	salt_str.	FP-1024 (394)	0.297 [.131,.450]	0.665	0.278	0.349	0.929
In vivo	XGB	salt_str.	FP-512 (266)	0.311 [−.014,.571]	0.860	0.237	0.319	0.983

PR-AUC baselines (prevalence): Ames 0.302, in vitro 0.124, in vivo 0.028.

Under fixed features and preprocessing (compact, raw SMILES), all seven algorithms on Ames spanned an MCC range of only 0.063 (Table 3, Figure 2); the five tabular models excluding LR spanned 0.023. Pairwise McNemar tests⁴³ between the top two models found no significant differences (Ames $p = 0.39$; in-vivo $p = 0.72$). This in-domain equivalence did not extend to cross-source settings (see Data Source Heterogeneity and Algorithm-Specific Robustness).

The GCN underperformed fingerprint-based approaches on Ames (mean MCC 0.402 across the three preprocessing scenarios, vs. 0.507 for XGB averaged across compact and FP-512/1024/2048 features under matched preprocessing); this is consistent with prior reports that graph networks need substantially larger datasets to outperform engineered features. The comparison should be read with the caveat that GCN was paired exclusively with graph features in this study, so the GCN–XGB gap conflates representation and algorithm.

Evaluation Cascade

For Ames, random-split CV reached $\text{MCC} = 0.670 \pm 0.010$, consistent with published benchmarks,^{13,15} while scaffold-split CV dropped to 0.624 ± 0.049 ($\Delta = 0.046$) and the held-out scaffold-split test to 0.470 (Table 4, Figure 3). Cross-source LDO produced a much larger collapse: drug $\rightarrow$ industrial reached $\text{MCC} = 0.234$, and industrial $\rightarrow$ drug reached $\text{MCC} = 0.122$. The data source heterogeneity gap ($\Delta = 0.390$) was an order of magnitude larger

Table 3: All-algorithm comparison (compact features, raw SMILES, Ames scaffold-split test). Range across all 7 algorithms: 0.063; range excluding LR and GCN: 0.023.

Algorithm	MCC	ROC-AUC
RF	0.492	0.838
ANN (MLP)	0.489	0.834
LGBM	0.479	0.833
SVM (RBF)	0.470	0.832
XGB	0.470	0.825
GCN	0.445	0.815
LR	0.430	0.800

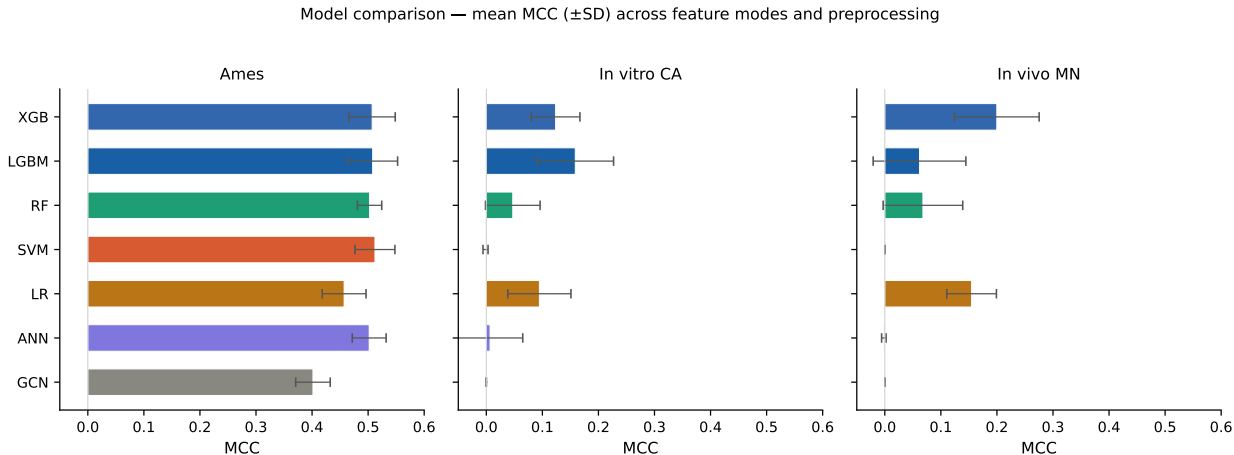


Figure 2: Mean MCC ($\pm$ SD) by algorithm across all feature modes and preprocessing scenarios, grouped by endpoint.

than the split-method gap ($\Delta = 0.046$). The LDO comparison involves additional structural factors—directional asymmetry, different test sizes, and different class distributions—so $\Delta = 0.390$ is best read as a combined effect, but the magnitude difference between the two effects is unambiguous. For in-vitro CA, random and scaffold splits gave nearly identical results (0.180 vs. 0.189); for in-vivo MN, both values were near random given their standard deviations.

Data Source Heterogeneity and Algorithm-Specific Robustness

Drug-sourced compounds ($n = 6,416$, positive rate 58.3%) and industrial-sourced compounds ($n = 7,144$, positive rate 3.0%) differed by a factor of 19.5 in mutagenicity prevalence (Fig-

Table 4: Evaluation cascade (locked configuration: XGBoost, compact features, raw SMILES). Level 2: mean $\pm$ SD (25 folds). Level 1: MCC [95% bootstrap CI]. Level 3: Ames only.

Level	Evaluation method	Ames MCC	In vitro MCC	In vivo MCC
2a	Random-split 5 $\times$ 5 CV	0.670 $\pm$ 0.010	0.180 $\pm$ 0.053	0.181 $\pm$ 0.094
2b	Scaffold-split 5 $\times$ 5 CV	0.624 $\pm$ 0.049	0.189 $\pm$ 0.084	0.104 $\pm$ 0.111
1	Scaffold-split test	0.470 [.434,.502]	0.114 [−.023,.270]	0.071 [−.029,.285]
3a	LDO: drug $\rightarrow$ industrial	0.234	—	—
3b	LDO: industrial $\rightarrow$ drug	0.122	—	—

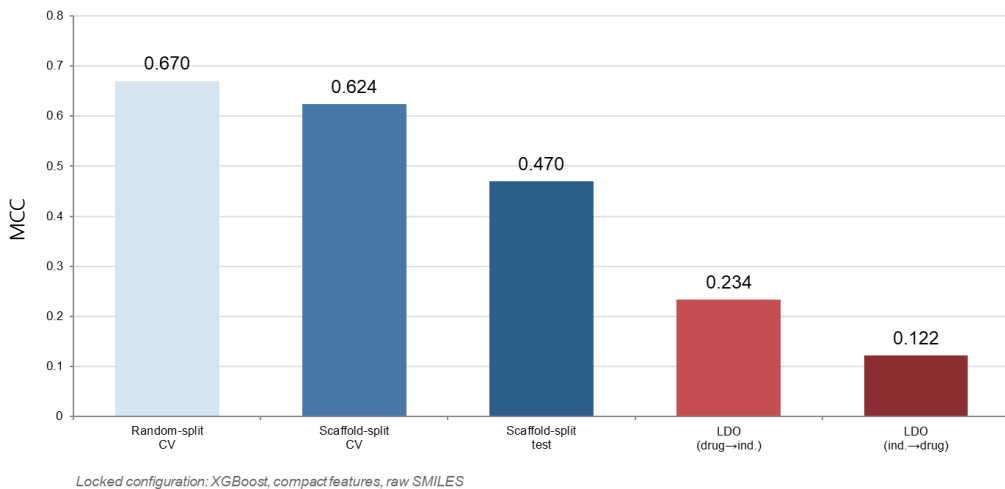


Figure 3: Evaluation cascade for Ames mutagenicity (locked configuration). The data source heterogeneity effect ($\Delta = 0.390$) substantially exceeds the split-method effect ($\Delta = 0.046$).

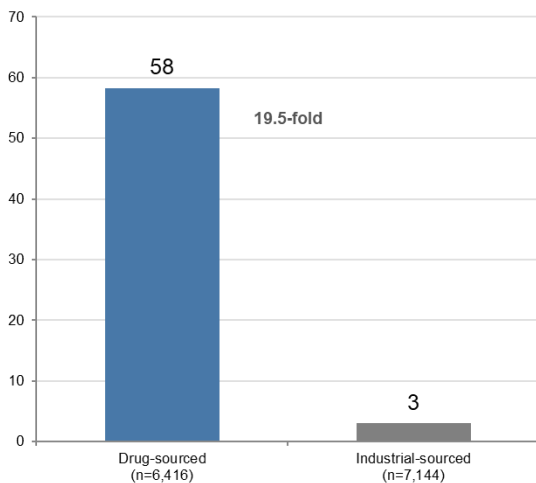
ure 4). A naive classifier predicting positive for drug-sourced and negative for industrial-sourced compounds achieved $\text{MCC} = 0.608$ and balanced accuracy = 0.834 without any structural information—comparable to the random-split CV MCC of 0.670 and within 0.062 MCC of the best in-domain model. The implication is that on mixed-source benchmarks a substantial fraction of apparent “predictive performance” may reflect the model’s ability to learn source membership—which is a property of how the benchmark was assembled—rather than mutagenicity per se. Published Ames model performance on similarly composed datasets should therefore be interpreted with caution unless cross-source evaluation is reported.

LDO with three algorithms (Table 5) revealed two patterns. Drug $\rightarrow$ industrial generalization was uniform across algorithms (MCC 0.232–0.235). In contrast, industrial $\rightarrow$ drug exposed sharp differences: XGB reached only MCC = 0.122 (sensitivity 0.062), LGBM 0.134, while RF maintained MCC = 0.325 (sensitivity 0.466). We hypothesize that RF’s better cross-source robustness reflects random feature subsampling at each split (random subspace method),³⁴ which reduces reliance on source-correlated feature subsets.

Table 5: Leave-domain-out results by algorithm (Ames). Drug: $n = 6,416$, pos. rate 58.3%. Industrial: $n = 7,144$, pos. rate 3.0%. Naive source classifier: MCC = 0.608. MCC values are reported with 95% bootstrap confidence intervals (500 iterations); the underlying per-row metrics including ROC-AUC, balanced accuracy, sensitivity, specificity, and Brier score are provided in Table S9 (`ames_leave_domain_out.csv`).

Algorithm	Drug $\rightarrow$ Ind. MCC [95% CI]	Ind. $\rightarrow$ Drug MCC [95% CI]	Ind. $\rightarrow$ Drug Sens.
XGB	0.234 [0.201, 0.264]	0.122 [0.103, 0.140]	0.062
LGBM	0.232 [0.199, 0.262]	0.134 [0.115, 0.151]	0.081
RF	0.236 [0.202, 0.267]	0.325 [0.304, 0.346]	0.466

(A) Source-domain composition



Naive source-domain classifier: MCC = 0.608, balanced accuracy = 0.834

(B) Leave-domain-out by algorithm

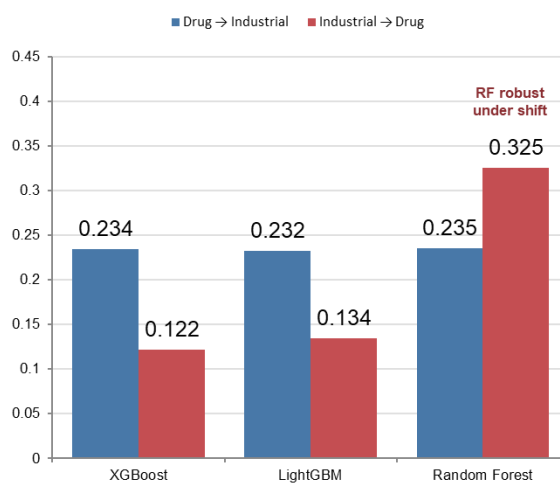


Figure 4: Data source heterogeneity in Ames. (A) Positive rate by source: drug 58.3% vs. industrial 3.0% (19.5-fold). (B) LDO MCC by algorithm; RF is substantially more robust in the industrial $\rightarrow$ drug direction.

Preprocessing, In-Vivo Behavior, and Feature Importance

Preprocessing. Salt stripping was strongly endpoint-specific (Figure 5). For in-vivo MN, the best salt-stripped model reached $\text{MCC} = 0.311$ versus 0.086 for the best raw model ($+0.225$); salt fragments were present in 83% of in-vivo SMILES, and among salt-containing compounds the positive rate was 2.8% versus 7.1% for non-salt compounds, suggesting salts were disproportionately attached to non-mutagenic compounds. For Ames, salt stripping had mixed effects ($\Delta = -0.03$ to $+0.02$); for in-vitro CA it was generally detrimental (mean $\text{MCC} 0.081 \rightarrow 0.056$). This argues against uniform preprocessing pipelines²⁰ and in favor of per-endpoint optimization.

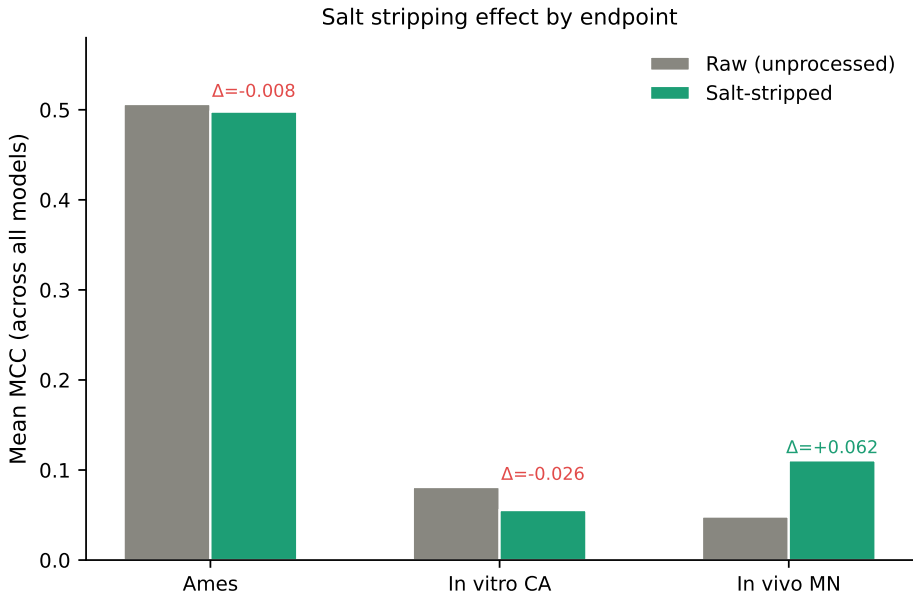


Figure 5: Effect of salt stripping on mean MCC (averaged across all models and feature modes). In-vivo improved by $+0.063$ on average and $+0.225$ for the best model; Ames mixed; in-vitro decreased.

In-vitro chromosome aberration. In-vitro CA performance was uniformly low across the cascade (random-split CV 0.180 ± 0.053 , scaffold-split CV 0.189 ± 0.084 , held-out test 0.114 with 95% CI crossing zero). Best-grid $\text{MCC} = 0.297$ (LGBM, salt-stripped, FP-1024) is substantially above the held-out CI but should be read as the upper envelope of search rather than a generalization estimate. Unlike in-vivo MN, the random-vs-scaffold gap was negligible

($\Delta = -0.009$), suggesting that in-vitro CA performance is not scaffold-leakage-limited but rather endpoint-limited under 2D descriptors. The class distribution is less extreme than in-vivo (10.4% vs. 3.7% positive rate), so test-set sample size is a smaller contributor to the wide CI than for in-vivo.

In-vivo micronucleus. The best in-vivo model achieved ROC-AUC = 0.860 and PR-AUC = 0.237 (an 8.5-fold enrichment over the 0.028 baseline); MCC = 0.311 with a 95% bootstrap CI of -0.014 to 0.571 reflects a combination of limited classification reliability at any fixed threshold *and* limited statistical power, since the test set contains only 12 positives among 431 compounds. In particular, the wide CI does not by itself license the strong claim that 2D descriptors are adequate or inadequate for binary classification of in-vivo MN; a larger, externally curated test set would be required to discriminate the two. The locked-configuration learning curve was flat at $\text{MCC} \approx 0.055$ from 10% to 100% of training data (Figure 7), though this curve was not computed for the salt-stripped configuration where the best-model improvement was observed.

Feature importance. SHAP analysis⁴² revealed consistent patterns (Figure 6): physicochemical descriptors dominated the top-5 (alert count and MW for Ames; LogP for in-vitro; MW for in-vivo), and no individual fingerprint bit appeared in any top-5 list despite fingerprints comprising 50–75% of features. Across all Ames models and scenarios, compact features yielded mean $\text{MCC} = 0.448$ versus 0.523 for FP-2048 ($\Delta = +0.075$), indicating that fingerprints add distributed but individually weak signal.

Learning curves. Locked-configuration learning curves (Figure 7) showed three patterns: Ames MCC saturated early (0.543 at 10%, 0.640 at 100%); in-vitro showed a noisy upward trend ($0.070 \rightarrow 0.170$); in-vivo was flat (≈ 0.055).

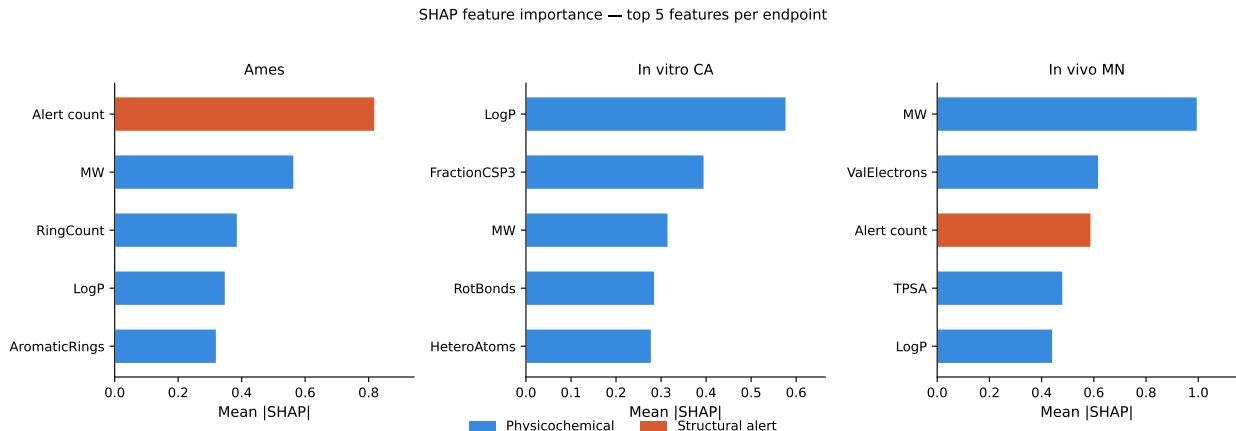


Figure 6: SHAP top-5 features per primary endpoint. Physicochemical descriptors dominate; structural alert count ranks first for Ames. No fingerprint bits appear in any top-5.

Discussion

The Performance Variation Hierarchy

The locked-configuration analysis yielded the hierarchy

$$\text{Data source heterogeneity} \gg \text{Splitting method} \approx \text{In-domain algorithm choice}$$

with the data source heterogeneity gap ($\Delta = 0.390$) an order of magnitude larger than the split-method gap ($\Delta = 0.046$), itself comparable to the in-domain algorithm spread (MCC range = 0.063). The hierarchy was established on a single Ames benchmark; the LDO Δ also reflects asymmetric test sizes and class distributions and should be read as an upper bound; and algorithm robustness was uneven under source shift, so algorithm choice may matter much more in cross-source deployment than the in-domain comparison suggests.

Importantly, the LDO experiment is not a deployment scenario. In practice, models are trained on all available data regardless of provenance, and the genotoxic potential of a chemical is fundamentally a property of its molecular structure, not of which database it was registered in. The point of cross-source evaluation is rather diagnostic: a large LDO drop reveals that apparent in-domain performance depends on the joint composition of training and test sets rather than on universally learned structure-activity relationships. This mat-

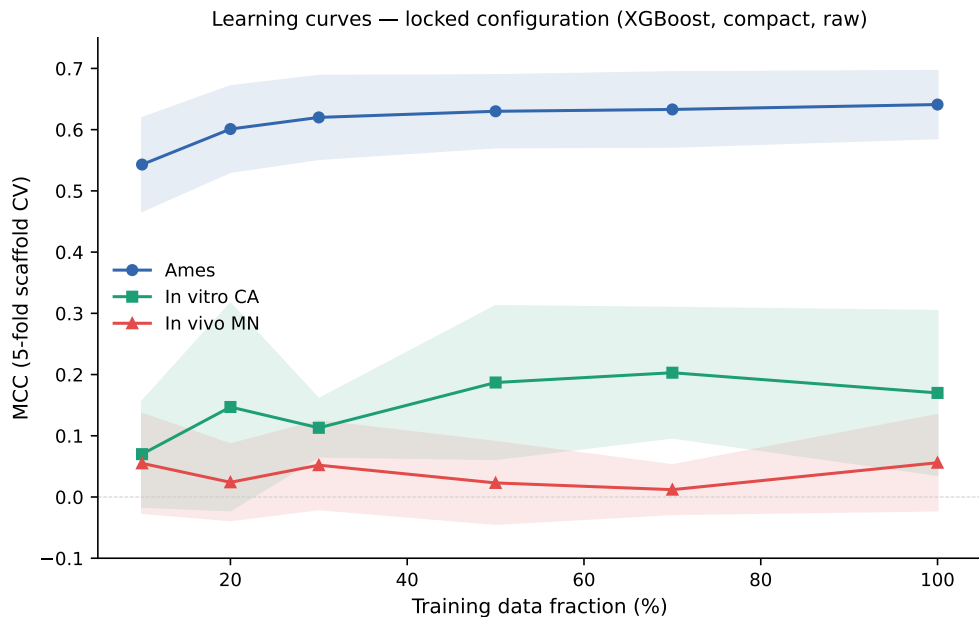


Figure 7: Learning curves (locked configuration). Shaded regions: ± 1 SD across 5-fold scaffold CV.

ters because new chemicals registered under EU REACH or K-REACH may occupy regions of chemical space that are sparsely represented in pharmaceutical benchmarks, and a benchmark MCC of 0.67 on a mixed-source test set provides little guarantee about the model’s reliability on such compounds. Cross-source LDO acts as a stress test that exposes this dependency and gives a more honest lower bound on out-of-distribution performance.

What Drives the Data Source Heterogeneity Gap?

Four non-exclusive hypotheses can be identified, in roughly decreasing order of parsimony. **Prior shift** from the 19.5-fold prevalence gap can mechanically lower MCC by miscalibrating the default threshold, consistent with XGB reaching sensitivity 0.062 in the industrial→drug direction. **Benchmark enrichment** is the most direct explanation for the prevalence gap itself: published Ames benchmarks such as Hansen¹³ were assembled to include a balanced set of known mutagens and non-mutagens for evaluation purposes, whereas industrial REACH/K-REACH dossiers reflect the prevalence of mutagenicity in the broader chemical

economy, which is genuinely low. **Data quality and labeling heterogeneity**—differing protocols, strain panels, S9 mix conditions, and curation conventions between drug and industrial sources—can shift both prevalence and decision boundaries. **Chemical space difference** would mean that even a perfectly calibrated cross-source model generalizes poorly because the two sources occupy systematically different regions of descriptor space; published QSAR-curation work documents the broad consequences of such differences for model transportability.^{18,20}

To estimate the contribution of prior shift, each LDO direction was evaluated at three thresholds: *default* ($t = 0.5$), *prior-corrected* ($t =$ training positive rate, optimal under pure prior shift), and *oracle* (t maximizing MCC on the test set, an upper bound on threshold-based recalibration). Two complementary analyses are provided: Table S7 uses Morgan FP-1024 + 13 physicochemical descriptors on the full LDO partition, and Table S8 uses the locked-configuration 138-feature compact set on a reduced cross-source partition (drug train $n = 5,027$; industrial train $n = 5,821$).

The headline quantitative result comes from Table S8: across all six algorithm-by-direction combinations on the locked-configuration features, threshold tuning explains 30.9–35.3 % of the total LDO gap (default-threshold MCC vs. in-domain scaffold-CV MCC of 0.6241), leaving a residual of $\Delta_{\text{residual}} = 0.15\text{--}0.37$ MCC that cannot be recovered by recalibration. The drug $\rightarrow$ industrial direction has a larger residual ($\Delta_{\text{residual}} = 0.34\text{--}0.37$) than industrial $\rightarrow$ drug ($\Delta_{\text{residual}} = 0.15\text{--}0.18$), consistent with industrial-to-drug generalization being supported by a larger and more diverse training set. The Table S7 analysis on the full partition with FP-1024 features shows the same qualitative pattern (prior-shift correction yields modest improvements, oracle MCC plateaus well below in-domain performance), confirming that the gap-decomposition conclusion is robust to feature choice.

In summary, prior shift accounts for roughly one-third of the LDO collapse, and the remaining two-thirds reflect genuine source shift—some combination of chemical space difference, data quality and labeling heterogeneity, and benchmark enrichment—that threshold

tuning alone cannot eliminate. Published Ames QSAR performance is therefore highly sensitive to source composition, and threshold tuning is not a sufficient remedy.

A note on the RF robustness finding. The default-threshold MCC values for RF differ across analyses: Table 5 (full partition, compact features) shows RF industrial $\rightarrow$ drug = 0.325, Table S7 (full partition, FP-1024 features) shows 0.102, and Table S8 (reduced partition, compact features) shows 0.350. Tables 5 and S8 are broadly consistent (difference ≤ 0.03), supporting the main-text claim that RF retains useful cross-source classification ability in the harder direction when run with domain-knowledge features. The Table S7 result on FP-1024 features differs more substantially, suggesting that the structural-alert component of the compact set carries non-trivial cross-source information that fingerprints do not fully replicate. The algorithm-comparison pattern at default threshold should therefore be read as feature-conditional. The within-row gap decomposition in Table S8 does not depend on algorithm choice in this way: all three algorithms show the same 30.9–35.3 % prior-shift share.

In-Vivo Micronucleus and Tiered Testing

The dissociation between ranking ability (ROC-AUC = 0.860, PR-AUC = 0.237) and classification reliability (MCC CI crossing zero) means that, on the present test set, in-vivo MN QSAR with 2D descriptors does not provide stable binary classification at any fixed threshold; a wider external evaluation would be needed to determine whether this reflects an intrinsic limit of 2D representations or a sample-size limit (only 12 positives in the test set). The 8.5-fold PR-AUC enrichment over baseline suggests value as one input in tiered screening under ICH M7⁷ or REACH,^{4,44} provided downstream decisions do not rely on threshold-based classification alone. The strong preprocessing sensitivity (+0.22 MCC from salt stripping) means reported in-vivo performance depends critically on preprocessing choices that are often undocumented.²¹ Mechanistically, micronucleus induction depends on ADME, metabolic activation, tissue distribution, and DNA repair,^{45,46} none of which 2D descriptors capture.

Cross-Endpoint Overlap and Preprocessing

The high compound overlap between endpoints (>90% shared with Ames) implies that performance differences (Ames MCC = 0.555 vs. in-vivo 0.311) reflect endpoint predictability from 2D structure rather than differing chemical spaces.¹⁸ Multi-task learning may be a productive direction.⁴⁷ Salt stripping was beneficial for in-vivo, mixed for Ames, and harmful for in-vitro—possibly because some counterions are themselves clastogenically relevant. The QSAR curation literature^{20,21} emphasizes standardization, but the present results suggest the optimal preprocessing strategy is endpoint-dependent.

Practical Recommendations

Based on these findings, we propose the following as useful additions to current reporting practices for genotoxicity QSAR (rather than as a fixed standard from a single study): (i) scaffold-based splitting; (ii) when source labels exist, cross-source LDO performance with disclosed source composition; (iii) per-endpoint preprocessing optimization, with preprocessing flags reported alongside metrics; (iv) bootstrap CIs and AD coverage; (v) for class-imbalanced endpoints, both threshold-free (ROC-AUC, PR-AUC) and threshold-dependent (MCC, balanced accuracy) metrics; (vi) testing algorithm superiority claims under source shift, not just in-domain—in our data, in-domain MCC range across seven algorithms was 0.063 while cross-source MCC range across three tree-based algorithms reached 0.203.

Several constraints bound these findings. A single scaffold split was used rather than repeated holdout, the GCN did not include edge features or pre-training,⁴⁸ and 3D or QM descriptors, metabolite predictions, within-AD vs. outside-AD stratification for LDO, and multiple-comparison correction for the McNemar tests²⁴ were not evaluated. The Klimisch quality filter was applied only to the industrial-sourced data, since the drug-sourced benchmarks do not carry equivalent annotations; this asymmetry is a residual confound for the cross-source comparison. Finally, the threshold-decomposition analysis was run with a different feature set than the main pipeline, so its absolute MCC values are not directly

comparable to Table 5 (see Table S7 footnote and the “note on the RF robustness finding” above).

Conclusion

An evaluation cascade applied to 225 model configurations across three genotoxicity datasets shows that, for the Ames mutagenicity benchmark studied here, data source heterogeneity—the mixing of drug-like and industrial compounds with sharply different prevalence and provenance—drives a substantially larger performance gap ($\Delta = 0.390$, scaffold CV $\rightarrow$ LDO) than the choice between random and scaffold splitting ($\Delta = 0.046$). A naive source classifier using no structural information reaches $\text{MCC} = 0.608$ on the same benchmark, suggesting that mixed-source Ames performance partly reflects learned source membership rather than mutagenicity. Threshold-tuning analysis attributes roughly one-third (30.9–35.3 %) of the LDO gap to prior shift, leaving a residual that no recalibration can fix. Algorithm differences are small in-domain ($\text{MCC range} = 0.063$) but become pronounced under source shift, with RF more robust than gradient boosting in the harder direction; this cross-source advantage is feature-conditional and warrants further confirmation. In-vivo micronucleus models show ranking signal (PR-AUC 8.5-fold above baseline) but classification metrics whose CIs do not exclude zero on the present test set. These findings motivate an evaluation cascade with source-awareness as a useful complement to existing practices for mutagenicity QSAR.

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Supporting Information

The Supporting Information is available free of charge. Summary tables are shown below; complete per-configuration data are provided as supplementary CSV files in the code repository.

Table S0. Endpoint classification criteria

The full mapping of OECD, EPA, and EU Test Guidelines to the three endpoint categories (Ames, in vitro CA, in vivo MN) used for curating all datasets is provided as `table_s0_endpoint_classification.xlsx`. This table was used to assign each raw study record to one of the target endpoints prior to any modeling.

Table S1. Random versus scaffold split comparison

Table 6: Random vs. scaffold split (XGBoost, compact features, raw SMILES, 5×5 CV).

Endpoint	n	Random MCC	Scaffold MCC	Δ
Ames	13,560	0.670 ± 0.010	0.624 ± 0.049	0.046
In vitro CA	1,619	0.180 ± 0.053	0.189 ± 0.084	-0.009
In vivo MN	2,153	0.181 ± 0.094	0.104 ± 0.111	0.077

Table S2. Full 225-experiment results

Complete metrics (MCC, bootstrap CI, ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, Brier score, AD coverage) for all 225 configurations are provided in `table_main_default_thresh`. The same archive includes `all_locked_test.csv` for the locked-configuration cascade and `scenario_cv_selection.csv` documenting the per-scenario CV-selection state.

Table S3. Hyperparameter tuning comparison

Default versus tuned MCC for all configurations, including optimized parameter values, are provided in `table_hp_tuning_comparison.csv`.

Table S4. Applicability domain coverage

Per-configuration AD coverage (Tanimoto ≥ 0.3 , 1024-bit Morgan FP) and mean similarity values are provided in `ad_coverage_summary.csv`. Coverage ranged from 48.5% (in vitro)

to 58.9% (in vivo) across configurations.

Table S5. Cross-endpoint compound overlap

Pairwise compound overlap and label concordance (κ) between the three endpoint datasets are provided in `cross_endpoint_overlap.csv`. Overlap with Ames was 97.5% for in vitro and 92.1% for in vivo.

Table S6. SHAP feature importance

SHAP values (mean |SHAP|, TreeExplainer) for all features across all tree-based models are provided in `shap_all_experiments.csv`; per-endpoint top-20 summaries are in `shap_top20_ames.csv`, `shap_top20_invitro.csv`, and `shap_top20_invivo.csv`. Statistical comparison results between top algorithms (McNemar, Cochran’s Q where applicable) are provided in `statistical_comparison_{ames,invitro,invivo,invitro_sampling,invivo_sampling}` and the cross-endpoint summary in `mcnemar_tests.csv`. Learning-curve raw data (Figure 7) are in `learning_curves.csv`.

Table S7. LDO threshold-tuning analysis (prior shift vs. source shift)

To distinguish prior shift (calibration mismatch from differing prevalence) from genuine source shift, we evaluated each LDO direction at three decision thresholds: default ($t = 0.5$), prior-corrected ($t =$ training positive rate), and oracle (t maximizing MCC on the test set, providing an upper bound). Per-direction and per-algorithm results are shown below.

Note on features and main-pipeline comparability. For computational tractability this analysis used Morgan FP-1024 + 13 physicochemical descriptors rather than the full 138-feature compact set used in the main pipeline. Absolute MCC values therefore differ from Table 5 (notably RF industrial→drug: 0.102 here vs. 0.325 in the main analysis). The internal three-threshold comparison within each row is valid because identical features are

used across thresholds, but the algorithm ranking observed at the default threshold should not be cross-referenced with Table 5, which uses a different feature set. Re-running this decomposition with the 138-feature compact set is identified in the Discussion as the highest-priority follow-up experiment.

Table 7: LDO threshold-tuning analysis (Morgan FP-1024 + 13 physchem features). Default = MCC at $t = 0.5$. Prior-corrected = MCC at t equal to the training positive rate. Oracle = upper bound MCC across all thresholds. The gap between oracle LDO MCC and the in-domain scaffold-CV MCC (≈ 0.624) isolates the irreducible source shift that cannot be corrected by threshold tuning.

Direction	Algo.	MCC default	MCC prior-corr.	MCC oracle	Oracle threshold
Drug $\rightarrow$ Ind.	XGBoost	0.231	0.263	0.310	0.788
Drug $\rightarrow$ Ind.	LightGBM	0.241	0.264	0.304	0.781
Drug $\rightarrow$ Ind.	RF	0.250	0.299	0.359	0.690
Ind. $\rightarrow$ Drug	XGBoost	0.200	0.267	0.273	0.041
Ind. $\rightarrow$ Drug	LightGBM	0.239	0.214	0.267	0.180
Ind. $\rightarrow$ Drug	RF	0.102	0.341	0.345	0.037

Mean improvement (default $\rightarrow$ prior-corrected): +0.075 (range -0.024 to $+0.239$). Mean improvement (default $\rightarrow$ oracle): +0.108 (range $+0.028$ to $+0.243$). Oracle MCC range: 0.267 – 0.359 , all substantially below the in-domain scaffold-CV MCC of 0.624 .

Table S8. LDO threshold-tuning analysis on the locked-configuration features (gap decomposition)

This table complements Table S7 by applying the same three-threshold decomposition with the locked-configuration features (138-feature compact: 13 physicochemical descriptors + 125 structural alerts on raw SMILES). Unlike Table 5, which uses the full within-domain partitions (drug $n = 6,416$; industrial $n = 7,144$), this analysis was performed on a reduced cross-source partition (drug train $n = 5,027$; industrial train $n = 5,821$); see methodological note below.

The shared in-domain reference is the locked-configuration scaffold-CV MCC (0.6241 , matching Table 4). The reported *plateau* interval is the oracle MCC ± 0.030 , intended as a rough robustness range under threshold perturbation, *not* a confidence interval.

Table 8: LDO threshold decomposition on the locked-configuration compact features. Δ_{total} = in-domain CV MCC – default-threshold LDO MCC. Δ_{prior} = oracle MCC – default LDO MCC (the part recoverable by threshold tuning). Δ_{residual} = in-domain CV MCC – oracle MCC (the irreducible source-shift component). Across all six rows, threshold tuning explains 30.9–35.3 % of the total LDO gap.

Direction	Algo.	MCC _{t=0.5}	Oracle t	Oracle MCC	Plateau	Δ_{total}	Δ_{prior}	Δ_{resid}	% expl.
Drug $\rightarrow$ Ind.	XGBoost	0.118	0.120	0.285	0.255–0.305	0.506	0.167	0.339	33.0
Drug $\rightarrow$ Ind.	LightGBM	0.103	0.120	0.270	0.240–0.290	0.521	0.167	0.354	32.0
Drug $\rightarrow$ Ind.	RF	0.089	0.120	0.254	0.224–0.274	0.535	0.165	0.370	30.9
Ind. $\rightarrow$ Drug	XGBoost	0.389	0.560	0.472	0.442–0.492	0.235	0.083	0.152	35.3
Ind. $\rightarrow$ Drug	LightGBM	0.372	0.560	0.458	0.428–0.478	0.252	0.086	0.166	34.1
Ind. $\rightarrow$ Drug	RF	0.350	0.560	0.440	0.410–0.460	0.274	0.090	0.184	32.8

In-domain scaffold-CV MCC = 0.6241 (locked configuration; shared across all rows). All MCC values rounded to 3 decimals from the source data. Test prior rates: drug = 0.59, industrial = 0.03.

Methodological note on partitioning. The reduced- N partition used in Table S8 reflects [TODO: confirm exact rule from pipeline—e.g., removal of cross-source duplicate compounds and/or scaffold-aware cross-domain partitioning before LDO]. As a result, the default-threshold MCCs here are not directly comparable to Table 5: in particular, the directional pattern at $t = 0.5$ is opposite (drug $\rightarrow$ industrial is harder here, industrial $\rightarrow$ drug harder in Table 5), reflecting the different test-set prevalence after partitioning. The within-row threshold comparison and the $\Delta_{\text{total}}/\Delta_{\text{prior}}/\Delta_{\text{residual}}$ decomposition remain internally valid because identical data and features are used across the three thresholds within each row.

Table S9. Leave-domain-out raw metrics with bootstrap confidence intervals (Ames, full partition)

Table S9 reports the complete per-row metrics underlying Table 5 on the full LDO partition (drug $n = 6,416$ pos. rate 58.3%; industrial $n = 7,144$ pos. rate 3.0%) at the default decision threshold. All metrics are computed with bootstrap 95% CIs (500 iterations); the underlying CSV is `ames_leave_domain_out.csv`.

The CI structure here is informative beyond the point estimates: the RF industrial $\rightarrow$ drug result (MCC 0.325 [0.304, 0.346]) is statistically clearly separated from XGB and LGBM

Table 9: LDO raw metrics with 95% bootstrap confidence intervals. MCC, balanced accuracy (BAcc), ROC-AUC, sensitivity (Sens), and specificity (Spec) are reported as point [lo, hi]. Brier score is the mean Brier loss; lower is better.

Direction	Algo.	MCC	BAcc	ROC-AUC	Sens	Spec
Drug $\rightarrow$ Ind.	XGB	0.234 [0.201, 0.264]	0.750 [0.718, 0.778]	0.813 [0.781, 0.842]	0.640 [0.577, 0.699]	0.859 [0.850, 0.867]
Drug $\rightarrow$ Ind.	LGBM	0.232 [0.199, 0.262]	0.753 [0.719, 0.784]	0.816 [0.785, 0.846]	0.655 [0.589, 0.717]	0.851 [0.843, 0.859]
Drug $\rightarrow$ Ind.	RF	0.236 [0.202, 0.267]	0.743 [0.711, 0.773]	0.816 [0.787, 0.843]	0.616 [0.552, 0.677]	0.871 [0.862, 0.879]
Ind. $\rightarrow$ Drug	XGB	0.122 [0.103, 0.140]	0.525 [0.520, 0.529]	0.739 [0.726, 0.752]	0.062 [0.054, 0.069]	0.988 [0.983, 0.992]
Ind. $\rightarrow$ Drug	LGBM	0.134 [0.115, 0.151]	0.531 [0.526, 0.536]	0.724 [0.713, 0.736]	0.081 [0.072, 0.090]	0.981 [0.976, 0.986]
Ind. $\rightarrow$ Drug	RF	0.325 [0.304, 0.346]	0.656 [0.646, 0.666]	0.737 [0.725, 0.749]	0.466 [0.450, 0.483]	0.846 [0.831, 0.860]

Brier scores (mean of bootstrap distribution): XGB drug $\rightarrow$ ind 0.124, ind $\rightarrow$ drug 0.456; LGBM drug $\rightarrow$ ind 0.127, ind $\rightarrow$ drug 0.464; RF drug $\rightarrow$ ind 0.117, ind $\rightarrow$ drug 0.241. The sharp Brier asymmetry between the two LDO directions reflects the prior-shift effect discussed in the main text and quantified in Table 8.

(MCC CIs upper-bounded at 0.140 and 0.151 respectively), supporting the claim that RF’s cross-source advantage in this direction is not attributable to point-estimate noise. By contrast, the three algorithms are statistically indistinguishable in the drug $\rightarrow$ industrial direction.

TOC Graphic

Some journals require a graphical entry for the Table of Contents. This should be laid out “print ready” so that the sizing of the text is correct. Inside the tocentry environment, the font used is Helvetica 8 pt, as required by *Journal of the American Chemical Society*.

The surrounding frame is 9 cm by 3.5 cm, which is the maximum permitted for *Journal of the American Chemical Society* graphical table of content entries. The box will not resize if the content is too big: instead it will overflow the edge of the box.

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